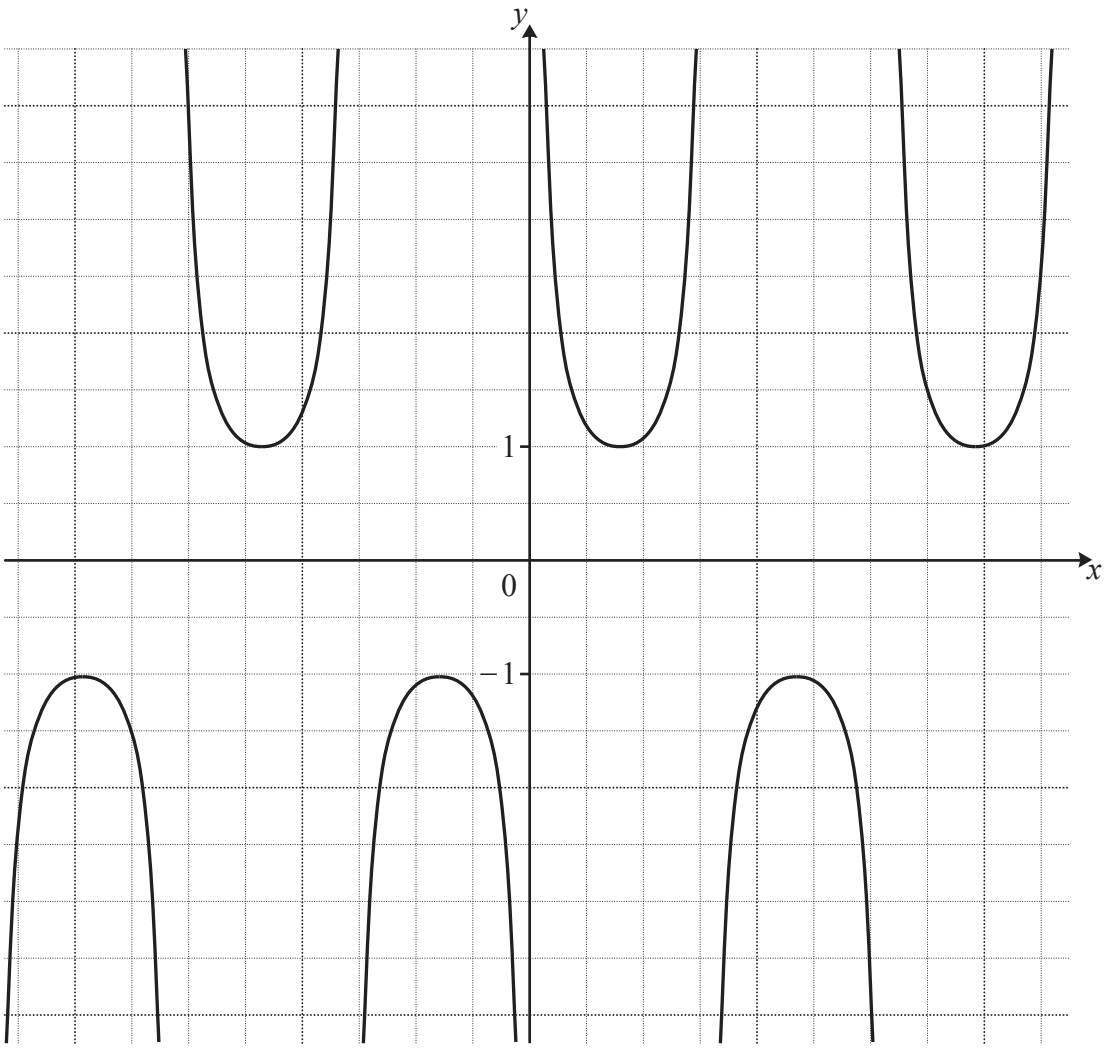
6 The diagram shows part of the graph of $f(x) = \text{cosec}\, 2x$.



(a) State which of the following is the equation of the curve $y = \dfrac{1}{f(x)}$.

$y = \cos 2x$ $y = \text{cosec}\,(-2x)$ $y = -\text{cosec}\, 2x$ $y = \sin 2x$ **[1]**

(b) State the exact equations of the asymptotes of $y = \text{cosec}\, 2x$ for $0 \leqslant x \leqslant \pi$. **[1]**